

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Assisted senior engineers with **site surveys** and **data collection** across industrial client locations, gathering critical information for energy assessments.
- Contributed to the analysis of **energy consumption data**, identifying key inefficiencies and supporting the development of **energy saving opportunities** for manufacturing clients.
- Supported the preparation of technical reports and **engineering calculations** for feasibility studies, directly aiding senior staff in client deliverables.
- Gained hands-on exposure to core industrial utility systems including **HVAC**, **Steam**, and **Compressed Air**, understanding their operational parameters.
- Participated in **energy audits** and baseline development, aligning with industry standards such as **IS 399** and **EN 16247**.
- Attended client meetings and site visits, building confidence in **technical communication** and stakeholder engagement with energy managers.

SKILLS

Engineering Disciplines – Mechanical Engineering, Electrical Engineering, Energy Engineering, Building Services Engineering

Energy & Sustainability – Industrial Decarbonisation, Carbon Intelligence, Energy Assessments, Site Surveys, Energy Audits, Energy Saving Opportunities, IS 399, EN 16247, ISO 50001

Industrial Systems – HVAC, Steam, Chilled Water, LPHW, Compressed Air, Cooling Towers, Electrical Distribution

Data Analysis & Reporting – Energy Consumption Data Analysis, Technical Reports, Feasibility Studies, Engineering Calculations, Measurement Verification (MV)

Tools & Technologies – AutoCAD, Revit, MATLAB, Excel, Python, Power BI, Tableau

Client Engagement – Technical Communication, Stakeholder Engagement, Client Meetings, Site Visits

PROJECTS

Industrial Energy Efficiency Assessment (HCL Technologies)

- Conducted detailed site surveys and data collection for a manufacturing client's energy systems, including steam and compressed air usage, to identify potential efficiency improvements.
- Analyzed operational data using **Python** and **Excel** to pinpoint areas of high energy consumption, contributing to a report that identified opportunities for **15%** energy reduction.
- Assisted in the preparation of technical documentation and feasibility studies for proposed energy efficiency upgrades, including cost estimates for new equipment.

Building Energy Management System Analysis (MSc Capstone)

- Developed a data analysis framework using **Power BI** and **DAX** to visualize energy consumption patterns from a building management system dataset.
- Modeled energy usage across HVAC and lighting systems, identifying anomalies and contributing to a final report recommending operational adjustments to reduce peak load by approximately **8%**.
- Presented findings and interactive dashboard prototypes to academic stakeholders, demonstrating clear communication of complex data insights.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)